

Hyperplane Configurations, Subtractive Sets, their Groupoids, and their Blowups

Jake Kettinger

Colorado State University
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Definition

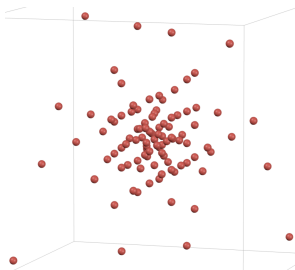
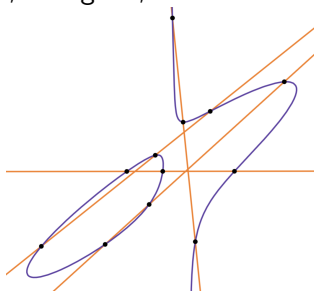
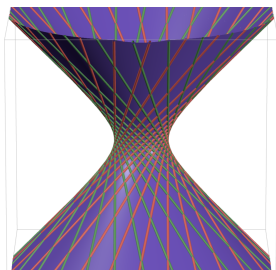
A finite set Z in $\mathbb{P}_k^n$ is **geproci** if the projection $\overline{Z}$ of Z from a general point P to a hyperplane $H = \mathbb{P}_k^{n-1}$ is a complete intersection in H .

Geproci stands for **g**eneral **p**rojection is a **c**omplete **i**ntersection. In characteristic 0, the only nontrivial examples known are for $n = 3$. In this case a hyperplane is a plane. A reduced set of points in a plane is a complete intersection if it is the transverse intersection of two algebraic curves.

For $\#Z = ab$ ($a \leq b$), Z is (a, b) -geproci if $\overline{Z}$ is the intersection of a degree a curve and a degree b curve.

Species of Geproci

Grids, half-grids, and non-half-grids



Background: k -Spreads in $\mathbb{P}_F^n$

Definition

Given a $k \geq 0$ and field F , a k -**spread** of $\mathbb{P}_F^n$ is a set of mutually-skew k -planes that partition $\mathbb{P}_F^n$.

Spreads are known to exist for $F =$ any finite field if and only if $k + 1$ divides $n + 1$, and also for $k + 1 = (n + 1)/2$ for $F = \mathbb{R}$.

Spreads are instrumental for the proof that $\mathbb{P}_{\mathbb{F}_q}^3$ is a $(q + 1, q^2 + 1)$ -geproci half-grid in $\mathbb{P}_{\mathbb{F}_q}^3$ [K– 24]. In this case, a spread is a partition of $\mathbb{P}_{\mathbb{F}_q}^3$ into lines.

Hopf Spreads

It is possible to construct a spread of k -dimensional hyperplanes inside $\mathbb{P}_{\mathbb{F}_q}^n$ if and only if $k + 1$ divides $n + 1$.

Let $p(x) \in \mathbb{F}_q[x]$ be irreducible, monic, and degree $k + 1$, and let P be its companion matrix. Then one can construct a spread of spaces of the form

$$\text{row} \left(\underbrace{0 \quad \cdots \quad 0 \quad \boxed{I_{k+1}} \quad A_1 \quad \cdots \quad A_j}_{\frac{n+1}{k+1}} \right) : A_i \in \mathbb{F}_q[P].$$

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For $A \in F^{(n+1) \times (n+1)}$, denote by $H(A)$ the rowspan of the matrix $\begin{pmatrix} I_{n+1} & A \end{pmatrix}$ as an n -plane in $\mathbb{P}_F^{2n+1}$, while $H_\infty := \text{rowsp} \begin{pmatrix} 0 & I_{n+1} \end{pmatrix}$.

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Maximal Partial Spreads

Note that a spread over $\mathbb{P}_{\mathbb{F}_q}^3$ comprises $q^2 + 1$ mutually-skew lines.

Definition

A **partial spread** of $\mathbb{P}_{\mathbb{F}_q}^3$ with **deficiency** d is a set of $q^2 + 1 - d$ mutually-skew lines. A **maximal partial spread** is a partial spread of positive deficiency that is not contained in any larger partial spread.

Maximal partial spreads are also instrumental to the study of geproci sets; in fact, given a maximal partial spread $\mathcal{M}$, the set $\mathbb{P}_{\mathbb{F}_q}^3 \setminus (\bigcup_{L \in \mathcal{M}} L)$ is geproci [K– 24].

Transversal-Mappings between n -planes

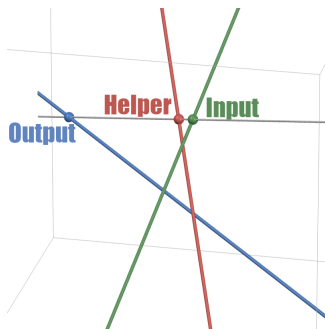
Definition

Given three n -planes $H_1, H_2, H_3 \subseteq \mathbb{P}_F^{2n+1}$ where $H_1 \cap H_2 = H_2 \cap H_3 = \emptyset$, we can define the function $\pi(H_1, H_2, H_3) : H_1 \rightarrow H_3$ as follows: take $p \in H_1$. Then there is a unique line T such that $p \in T$, $T \cap H_2 \neq \emptyset$, and $T \cap H_3 \neq \emptyset$. Then define $\pi(H_1, H_2, H_3)(p) = T \cap H_3$. This is the **projection** of H_1 to H_3 **via** H_2 .

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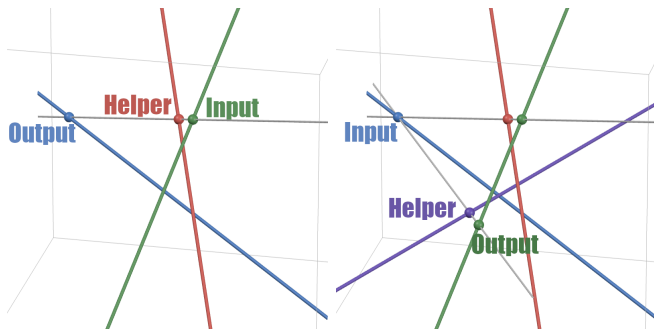
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Definition

A **groupoid** is a category $\mathcal{G}$ where every morphism is invertible.

- For any object $G \in \mathcal{G}$, $\text{Hom}_{\mathcal{G}}(G, G) = \text{Aut}_{\mathcal{G}}(G)$ is a group. $\text{Aut}_{\mathcal{G}}(G)$ is a “group of the groupoid.”
 - Whenever $\text{Hom}_{\mathcal{G}}(G_1, G_2) \neq \emptyset$, then $\text{Aut}_{\mathcal{G}}(G_1) \cong \text{Aut}_{\mathcal{G}}(G_2)$.
- So when $\text{Hom}_{\mathcal{G}}(G_1, G_2) \neq \emptyset$ for all $G_1, G_2 \in \mathcal{G}$, $\mathcal{G}$ induces only one group of the groupoid, up to isomorphism. Then it makes sense to say “the” group of the groupoid, $\text{Aut}_{\mathcal{G}}$.

Groupoids of n -planes

Theorem

Let $\mathcal{H}$ be a set of n -planes in $\mathbb{P}_F^{2n+1}$. Define Π to be the composition-closure of the set of functions $\{\pi(H_i, H_j, H_k) \mid H_i, H_j, H_k \in \mathcal{H}, H_i \cap H_j = H_j \cap H_k = \emptyset\}$. Then $(\mathcal{H}, \Pi)$ is a groupoid.

In this case, any group of the groupoid is a subgroup of $\text{Aut}(\mathbb{P}_F^n) \cong \text{PGL}(n+1, F)$.

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What can we say about this groupoid and its corresponding group(s)? In characteristic 0, when is it finite versus infinite? What is the relationship (if $\exists$) between $\text{Aut}_{(\mathcal{H}, \Pi)} \leq \text{Aut}(\mathbb{P}^n)$ and $\text{Aut}(\mathcal{H}) \leq \text{Aut}(\mathbb{P}^{2n+1})$? In characteristic 0 and $n = 1$, it is known that when $\text{Aut}_{(\mathcal{H}, \Pi)}$ is finite, then orbits of points under this groupoid are geproci [CFFHMSS 25+].

Ganger's Results

In her 2024 thesis, Ganger used the technique of **transversals** to prove the following theorem:

Theorem (Ganger Corollary 2.5)

The group of the groupoid for the spread induced by the degree-2 field extension $\mathbb{F}_{q^2}/\mathbb{F}_q$ over a finite field is isomorphic to the quotient $\mathbb{F}_{q^2}^/\mathbb{F}_q^* \cong C_{q+1}$.*

Definition

Given a set of lines $\mathcal{L}$ in $\mathbb{P}_F^3$, a **transversal** is a line T in $\mathbb{P}_F^3$ such that $T \cap \bar{L} \neq \emptyset$ for all $L \in \mathcal{L}$.

The Hopf spread has exactly two transversals T_1, T_2 for any finite field. The intersection of transversal with a line $L \in \mathcal{S}$ is a fixed point of $\text{Aut}_{(\mathcal{S}, \Pi)}(L)$!

Representing $\text{Aut}_{\mathcal{G}} \leq \text{PGL}(n+1, F)$

Let $H_1 = \langle u_i \rangle_{i=0}^n, H_2 = \langle v_i \rangle_{i=0}^n, H_3 = \langle w_i \rangle_{i=0}^n$ be n -planes in $\mathbb{P}^{2n+1}$. Then there is a unique matrix $A \in \text{PGL}(n+1)$ such that the following diagram commutes.

$$\begin{array}{ccc} \mathbb{P}^n & \xrightarrow{A} & \mathbb{P}^n \\ \downarrow \langle u_i \rangle & & \downarrow \langle w_i \rangle \\ H_1 & \xrightarrow{\pi(H_1, H_2, H_3)} & H_3 \end{array}$$

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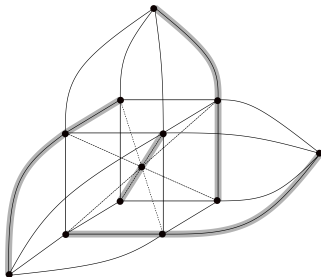
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By taking $H_i = H(M_i)$, A can be taken as $(M_3 - M_2)^{-1}(M_1 - M_2)$ [Favacchio 25+].

Example: The D_4 configuration

Theorem (K-)

Let $\mathcal{L}$ be the 16 lines of the $(12_4, 16_3)$ configuration D_4 and let Π be the composition-closure of the projection functions. Then the group of the groupoid $\mathcal{G} = (\mathcal{L}, \Pi)$ is $\text{Aut}_{\mathcal{G}} \cong D_6$.



In fact, one can find a subset of six lines whose group of the groupoid is D_6 [K- 25+]. Such a group is impossible to find in skew configurations [Favacchio 25+].

Finite Groupoids for $n = 1$

The finite subgroups of $\mathrm{PGL}(2, \mathbb{C})$ are C_n , D_{2n} , A_4 , S_4 , and A_5 . All of these except of the dihedral groups can be realized as the groupoid of a configuration of lines in $\mathbb{P}_{\mathbb{C}}^3$ [F 25+].

In characteristic $p > 0$, we can also attain groups $C_p \times C_p$, $C_p \rtimes C_{p-1}$, and D_{2n} for $p \mid n$.

Finite Groupoids for $n = 2$

In characteristic 0, the only known finite groups of the groupoid are cyclic. In these configurations, the orbit of a generic point is trivially geproci: the orbit restricted to each plane is a set of collinear points, and the whole orbit only spans a $\mathbb{P}^3$, forming a half-grid.

Blowing Up the n -planes

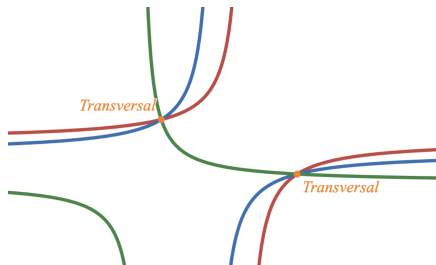
Consider the n -planes $U, V, W \subseteq \mathbb{P}^{2n+1}$, the blowup map $\beta : \text{Bl}_W(\mathbb{P}^{2n+1}) \rightarrow \mathbb{P}^{2n+1}$, and the exceptional locus $E = \beta^{-1}(W) \cong \mathbb{P}^n \times \mathbb{P}^n$. Then the projection $\pi : U \xrightarrow{V} W$ extends to a map $\tilde{\pi} : U \rightarrow E$. In the Chow ring $A(E) \cong \mathbb{Z}[\alpha, \beta]/(\alpha^{n+1}, \beta^{n+1})$,

$$[\tilde{\pi}(U)] = \sum_{i=0}^n \alpha^i \beta^{n-i}.$$

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Three n -planes $U, V, W \subseteq \mathbb{P}^{2n+1}$ uniquely determine a rational normal scroll $\mathbb{P}^1 \times \mathbb{P}^n \cong S \subseteq \mathbb{P}^{2n+1}$ (the Segre embedding, up to coordinates). Then $\tilde{\pi}(U) = \tilde{S} \cap E$.

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Example

Consider $\mathcal{L} = \left\{ H_\infty, H(0), H(I_2), H \begin{pmatrix} \zeta_6 & 1 \\ 0 & \zeta_6^5 \end{pmatrix}, H \begin{pmatrix} \zeta_6 & 0 \\ 1 & \zeta_6^5 \end{pmatrix} \right\}$ in $\mathbb{P}_{\mathbb{C}}^3$. Then $\text{Aut}_{\mathcal{L}} \cong A_4$, and the six induced bidegree-(1,1) curves in E_∞ have a stabilizer group $S_4 \rtimes C_2$.

The Conic Invariant

Let $H_0, \dots, H_4 \subseteq \mathbb{P}_F^5$, and let E be the exceptional locus after blowing up H_0 . Then the six pairs of the remaining planes each induce an $\alpha^2 + \alpha\beta + \beta^2$ -surface in E that meet each other in a $(12_3, 6_6)$ configuration.

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Set $A = (M_1 - M_2)^{-1}(M_2 - M_3)$ and $B = (M_1 - M_2)^{-1}(M_2 - M_4)$. Then the six surfaces satisfy the **conic condition** if and only if $\det(AB - BA) = 0$.